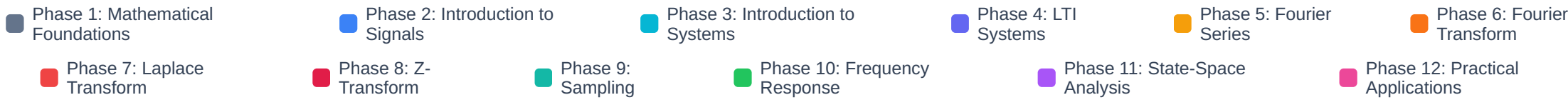


SIGNALS AND SYSTEMS

Complete Learning Roadmap — Beginner to Interview-Ready | 12 Phases · Prerequisite Flow · Milestones



PHASE 1 Mathematical Foundations

KEY CONCEPTS

- Complex numbers & Euler's identity $e^{j\theta}$
- Differentiation & integration rules
- Solving ordinary differential equations
- Matrix and linear algebra basics

FULL TOPIC LIST

Complex Numbers, Euler's Formula, Trigonometry, Exponentials, Logarithms, Differentiation, Integration, Partial Differentiation, ODEs, Matrix Basics

✓ **Learning Outcome / Milestone:** Solve the mathematical problems required for Signals and Systems.



PHASE 2 Introduction to Signals

↳ Builds on: function notation & graph-sketching from Phase 1

KEY CONCEPTS

- Continuous-time vs discrete-time signals
- Standard test signals: step, impulse, ramp, exponential, sinusoid
- Energy vs power signal classification
- Signal operations: shifting, scaling, reversal

FULL TOPIC LIST

CT/DT, Analog/Digital, Periodic/Aperiodic, Deterministic/Random, Energy/Power, Even/Odd signals; Standard signals (step, impulse, ramp, exponential, sinusoid, pulse); Operations (shift, scale, reversal)

✓ **Learning Outcome / Milestone:** Identify and classify different signal types.



PHASE 3 Introduction to Systems

↳ Requires: signal definitions from Phase 2

KEY CONCEPTS

- Linearity & time invariance (LTI test)
- Causality and stability conditions
- Memoryless vs dynamic systems
- Invertibility of a system

FULL TOPIC LIST

Input/output, CT/DT systems; Properties: linearity, time invariance, causality, stability, memory, invertibility

✓ **Learning Outcome / Milestone:** Analyze whether a system satisfies required properties.



PHASE 4 LTI Systems

↳ Requires: system properties (Phase 3) + integration (Phase 1)

KEY CONCEPTS

- Impulse response fully characterizes an LTI system
- Convolution integral / convolution sum
- Differential & difference equation models
- Step response from impulse response

FULL TOPIC LIST

LTI concept, impulse response, convolution, step response, differential & difference equation representation

✓ **Learning Outcome / Milestone:** Solve convolution problems and determine system outputs.



PHASE 5 Fourier Series

↳ Requires: LTI response to sinusoids (Phase 4)

KEY CONCEPTS

- Periodic signals as a sum of harmonics
- Trigonometric vs exponential Fourier series
- Frequency spectrum representation
- Parseval's theorem — energy in time vs frequency

FULL TOPIC LIST

Need for frequency analysis, orthogonality, trigonometric & exponential FS, harmonics, frequency spectrum, Parseval's theorem

✓ **Learning Outcome / Milestone:** Represent periodic signals using Fourier Series.





PHASE 6

Fourier Transform

↳ Math dependency: generalizes Fourier Series (Phase 5)

KEY CONCEPTS

- Extending Fourier Series to aperiodic signals
- CTFT and DTFT definitions
- Duality between time and frequency domains
- Frequency response of an LTI system

FULL TOPIC LIST

CTFT, DTFT, FT properties, time–frequency duality, frequency response, bandwidth



Learning Outcome / Milestone: Transform signals between time and frequency domains.



PHASE 7

Laplace Transform

↳ Math dependency: generalizes Fourier Transform (Phase 6) for CT systems

KEY CONCEPTS

- Laplace as a generalization of the Fourier Transform
- Region of Convergence (ROC) and its role in invertibility
- Transfer function $H(s)$ of a system
- Stability from pole locations in the s -plane

FULL TOPIC LIST

Bilateral & unilateral Laplace transform, ROC, properties, initial/final value theorems, transfer function; applications: solving ODEs, stability analysis



Learning Outcome / Milestone: Analyze continuous-time systems using Laplace Transform.



PHASE 8

Z-Transform

↳ Math dependency: discrete-time analogue of Laplace Transform (Phase 7)

KEY CONCEPTS

- Z-transform as the discrete-time counterpart of Laplace
- ROC and stability on the z -plane (unit circle)
- Pole–zero plots for digital filters
- Transfer function $H(z)$ of discrete systems

FULL TOPIC LIST

Introduction, ROC, properties, inverse Z-transform, transfer function, pole–zero plot, stability; applications: difference equations, digital systems



Learning Outcome / Milestone: Analyze discrete-time systems using Z-Transform.



PHASE 9

Sampling

↳ Bridges CT concepts (Phase 6/7) to DT concepts (Phase 8)

KEY CONCEPTS

- Sampling theorem & Nyquist rate
- Aliasing — causes and prevention
- Signal reconstruction from samples
- Basics of quantization

FULL TOPIC LIST

Sampling process, sampling theorem, Nyquist rate, aliasing, reconstruction, quantization basics



Learning Outcome / Milestone: Explain proper sampling and avoid aliasing.



PHASE 10

Frequency Response

↳ Requires: Fourier/Laplace/Z-domain analysis (Phases 6–8)

KEY CONCEPTS

- Magnitude and phase response of a system
- Bode plot construction basics
- Resonance conditions
- Filter types and their frequency behavior

FULL TOPIC LIST

Magnitude & phase response, Bode plots, resonance; filter types: low-pass, high-pass, band-pass, band-stop



Learning Outcome / Milestone: Analyze filter behavior using frequency response.





PHASE 11

State-Space Analysis

↳ Requires: matrix algebra (Phase 1) + differential equations (Phase 4)

KEY CONCEPTS

- State variables & state equations
- Matrix (A, B, C, D) system representation
- State transition matrix
- Controllability and observability

FULL TOPIC LIST

State variables, state equations, matrix representation, state transition matrix, controllability, observability



Learning Outcome / Milestone: Represent systems using state-space models.



PHASE 12

Practical Applications

↳ Integrates: all previous phases into applied engineering practice

KEY CONCEPTS

- Applying transforms to real audio/image/speech data
- Signal processing in biomedical & radar/sonar systems
- Role of S&S in control, embedded systems & IoT
- Vibration analysis in mechanical/robotic systems

FULL TOPIC LIST

Audio, image & speech processing, biomedical signals, radar, sonar, wireless communication, control systems, embedded systems, robotics, IoT, vibration analysis



Learning Outcome / Milestone: Connect theory with real-world engineering systems.

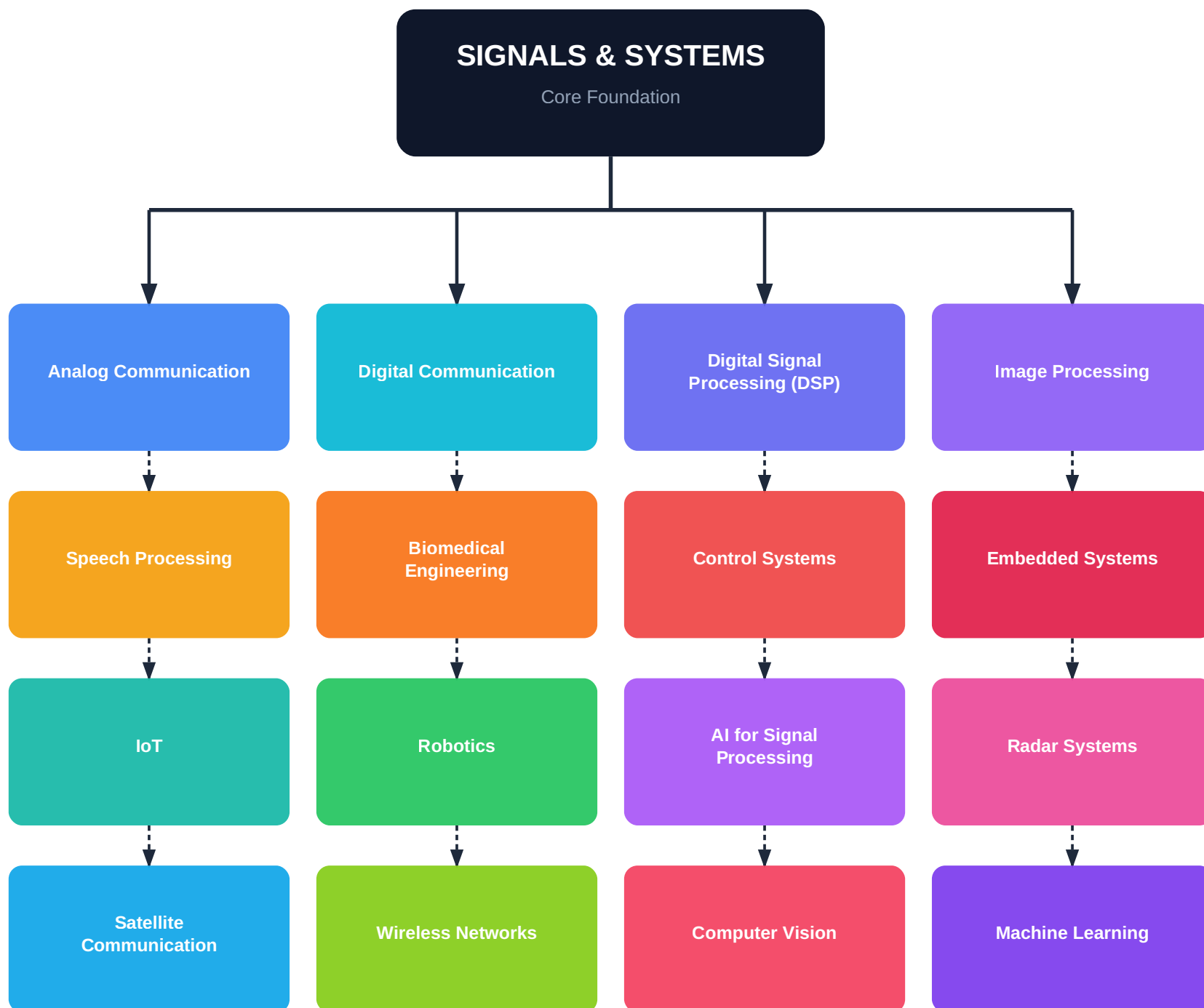
Mathematical Dependency Track

The core transform chain — each stage generalizes or specializes the previous one



Final Connections — Where Signals & Systems Leads

How this foundation branches into advanced ECE subjects and industry domains



The Learning Cycle

Repeat this 10-step workflow for every topic in the roadmap above

